Python List Methods

Complete Guide with Syntax & Examples

Starting list: `a = [1, 2, 3, 4, 5, 56]`

1. append()

Adds a single element to the END of the list.

Syntax: `list.append(element)`

```
a = [1, 2, 3, 4, 5, 56]
a.append(100)
print(a)
```

```
Output: [1, 2, 3, 4, 5, 56, 100]
```

2. clear()

Removes ALL elements from the list. Makes it empty.

Syntax: `list.clear()`

```
a = [1, 2, 3, 4, 5, 56]
a.clear()
print(a)
```

```
Output: []
```

3. copy()

Returns a SHALLOW COPY of the list. Changes to the copy do not affect the original.

Syntax: `new_list = list.copy()`

```
a = [1, 2, 3, 4, 5, 56]
b = a.copy()
b.append(999)
print(a) # original unchanged
print(b) # copy has new element
```

```
Output: [1, 2, 3, 4, 5, 56]
[1, 2, 3, 4, 5, 56, 999]
```

4. count()

Returns how many times a specific value appears in the list.

Syntax: `list.count(value)`

```
a = [1, 2, 3, 2, 2, 56]
print(a.count(2))
```

```
Output: 3
```

5. extend()

Adds ALL elements from another list (or iterable) to the end of the list.

Syntax: `list.extend(iterable)`

```
a = [1, 2, 3, 4, 5, 56]
a.extend([70, 80, 90])
print(a)
```

```
Output: [1, 2, 3, 4, 5, 56, 70, 80, 90]
```

6. index()

Returns the INDEX (position) of the FIRST occurrence of a value. Raises ValueError if not found.

```
Syntax: list.index(value, start, end) # start & end are optional
```

```
a = [1, 2, 3, 4, 5, 56]
print(a.index(4))
```

```
Output: 3
```

7. insert()

Inserts an element at a SPECIFIC POSITION (index) in the list.

```
Syntax: list.insert(index, element)
```

```
a = [1, 2, 3, 4, 5, 56]
a.insert(2, 99) # insert 99 at index 2
print(a)
```

```
Output: [1, 2, 99, 3, 4, 5, 56]
```

8. pop()

Removes and RETURNS the element at the given index. Default is the LAST element.

```
Syntax: list.pop(index) # index is optional, default = -1
```

```
a = [1, 2, 3, 4, 5, 56]
removed = a.pop() # removes last
print(removed)
print(a)
```

```
Output: 56
[1, 2, 3, 4, 5]
```

9. remove()

Removes the FIRST occurrence of a specific VALUE from the list. Raises ValueError if not found.

```
Syntax: list.remove(value)
```

```
a = [1, 2, 3, 4, 5, 56]
a.remove(3)
print(a)
```

```
Output: [1, 2, 4, 5, 56]
```

10. reverse()

Reverses the list IN-PLACE (modifies the original list, returns nothing).

```
Syntax: list.reverse()
```

```
a = [1, 2, 3, 4, 5, 56]
a.reverse()
print(a)
```

```
Output: [56, 5, 4, 3, 2, 1]
```

11. sort()

Sorts the list IN-PLACE in ascending order by default. Use reverse=True for descending.

```
Syntax: list.sort(key=None, reverse=False)
```

```
a = [5, 1, 56, 3, 2, 4]
a.sort() # ascending
print(a)
a.sort(reverse=True) # descending
print(a)
```

```
Output: [1, 2, 3, 4, 5, 56]
[56, 5, 4, 3, 2, 1]
```

Tip: append() adds ONE item | extend() adds MANY items | pop() returns the removed item | remove() removes by value